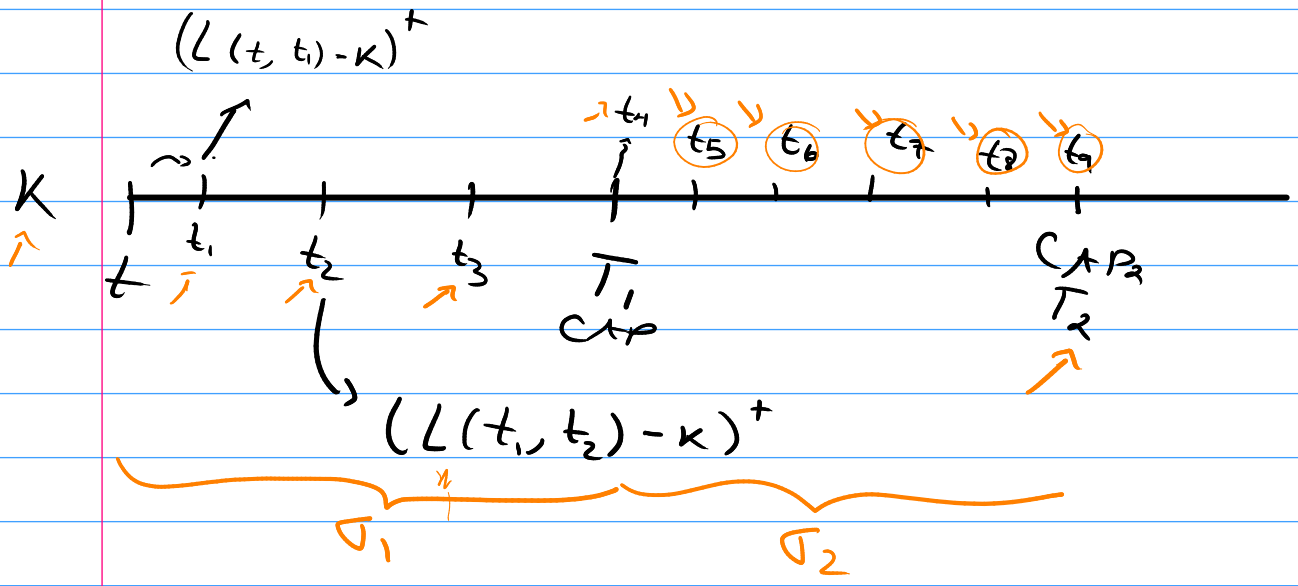
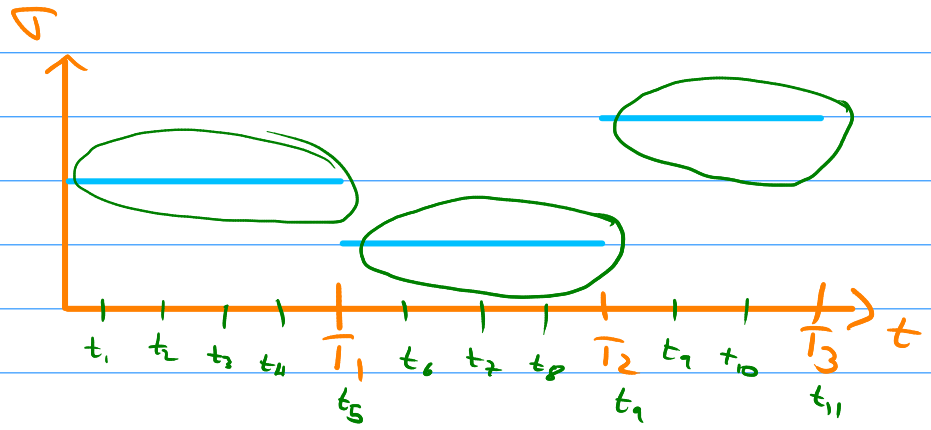
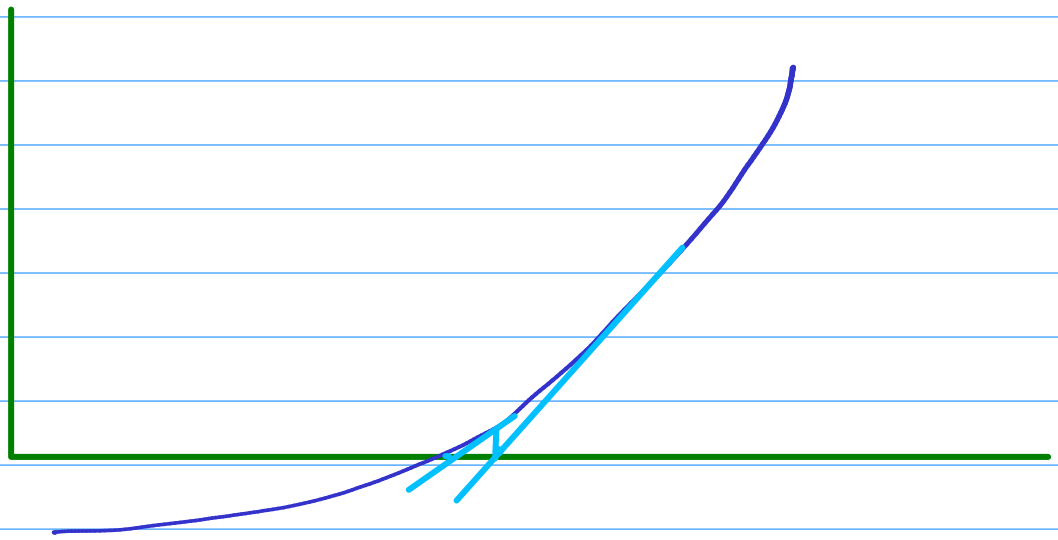
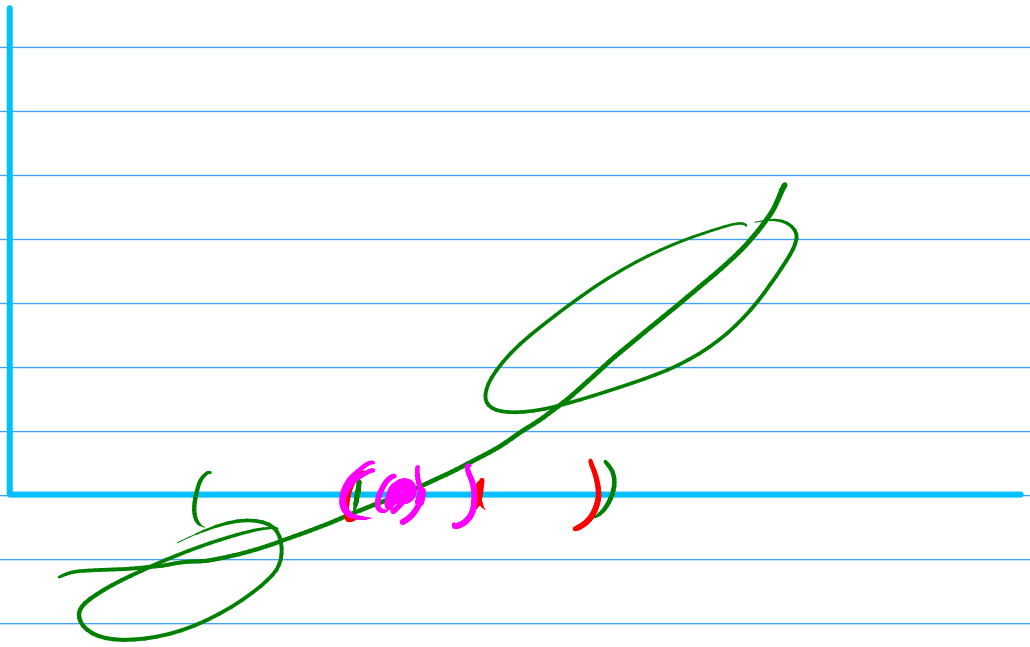
$K < L(t_i, t_{i+1})$ → El cap paga $L(t_i, t_{i+1})$

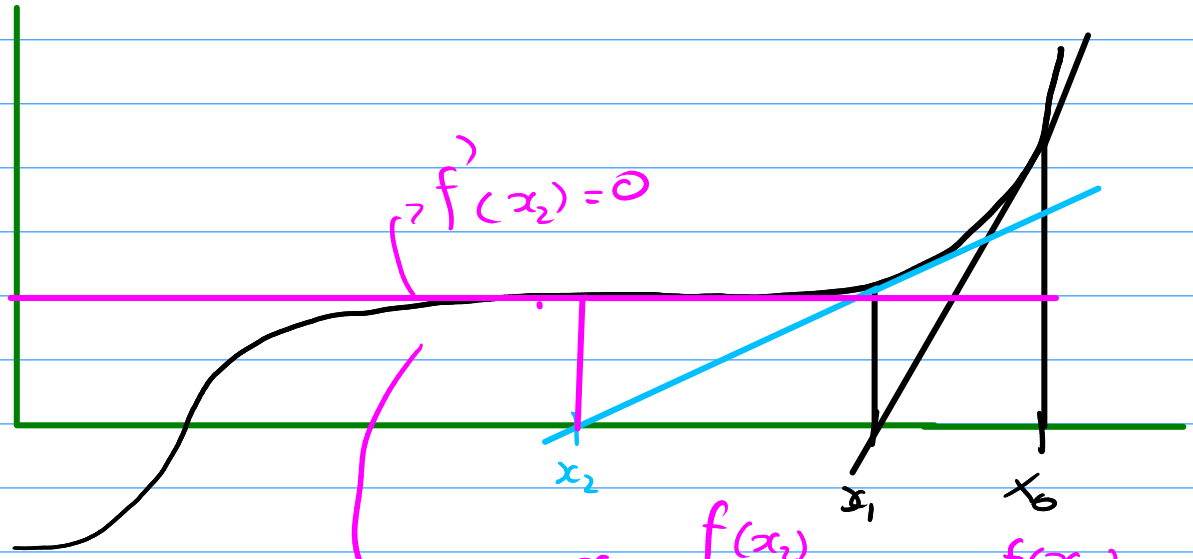
$K > L(t_i, t_{i+1})$ → No se paga nada









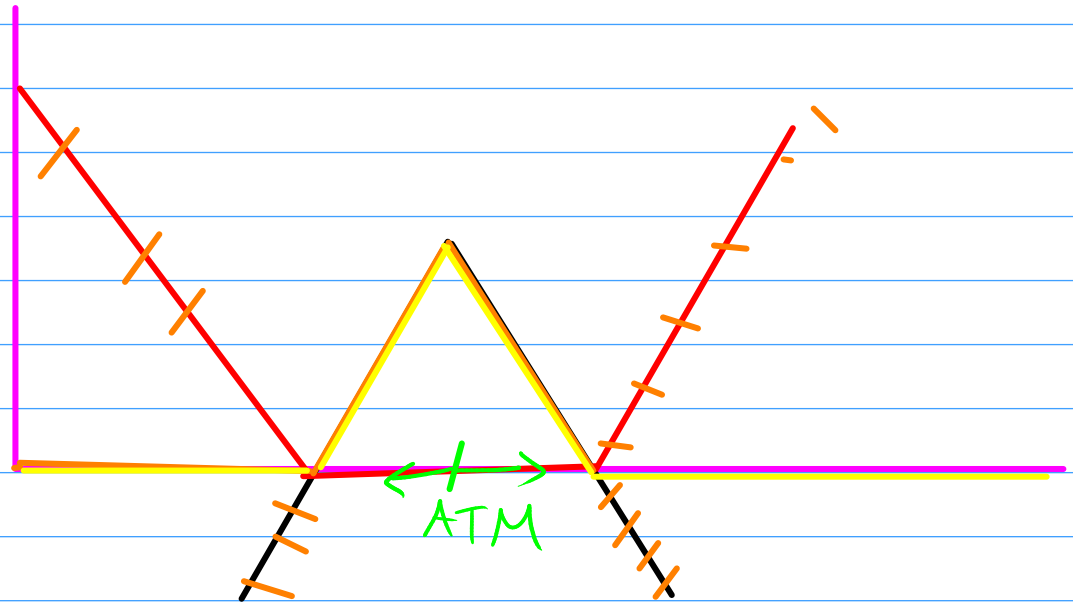


$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)} \rightarrow \frac{f(x_2)}{0}$$

Indeterminación!

$$\sqrt{\sum_{i=1}^{n} (\nabla_{mkt} - \nabla_{teo})^2}$$

$$\sqrt{\sum_{i=1}^{n} (cap(\nabla_{mkt}) - cap(\nabla_{teo}))^2}$$



$$FI_y = \sigma_{25\,C} - 2\sigma_{ATM} + \sigma_{25\,\Delta P}$$

$$d_1 = \frac{\ln\left(\frac{F}{K}\right) + \left(\frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}}$$

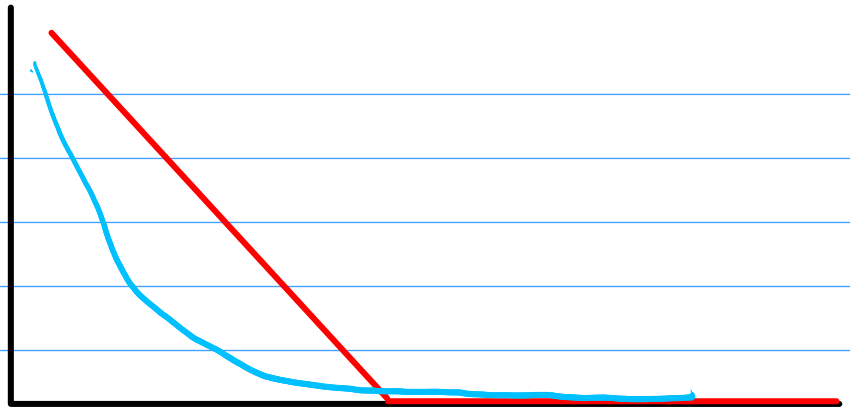
$\hookrightarrow d_1(\sigma)$

$$d_2 = d_1 - \sigma\sqrt{T}$$

$\hookrightarrow d_2(\sigma)$

$$\sigma_{25\Delta\,call} = 0.15 \quad , \quad F = 18.2 \quad , \quad T = 1y$$

$$0.25 = N(d_1) \Rightarrow 0.25 = N\left(\frac{\ln\left(\frac{18.2}{K}\right) + 0.15\left(\frac{1}{2}\right)}{0.15\sqrt{1}}\right)$$



$$\tilde{V}(S,\sigma) = V(S_0,\sigma_0) + \frac{\partial V}{\partial S}(S-S_0) + \frac{1}{2}\frac{\partial^2 V}{\partial S^2}(S-S_0)^2 \dots$$

$$+ \frac{\partial V}{\partial \sigma}(\sigma-\sigma_0) + \frac{1}{2}\frac{\partial^2 V}{\partial \sigma^2}(\sigma-\sigma_0)^2 + \dots$$

$$+ \frac{\partial^2 V}{\partial \sigma \partial S}(\sigma-\sigma_0)(S-S_0) \dots$$

$$f(x) = f(x_0) + f'(x_0)(x-x_0) + \frac{f''(x_0)}{2}(x-x_0)^2 + \dots$$

$$+ \frac{1}{n!} f^{(n)}(x_0)(x-x_0)^n + \dots$$

Definición: $V = \alpha_1 C_{25D} + \alpha_2 P_{25D} + \alpha_3 ATM$

$$\sigma_{ATM} = \sigma_{ATM} + 0\,\sigma_{25\,call} + 0\,\sigma_{25\,Put}$$

$$\sigma_{RR_{25D}} = 0 \cdot \sigma_{ATM} + 1\,\sigma_{25\,call} - 1\,\sigma_{25\,Put}$$

$$\sigma_{Fly} = -1\,\sigma_{ATM} + 0.5\,\sigma_{25\,call} + 0.5\,\sigma_{25\,Put}$$

$$\begin{pmatrix} \sigma_{ATM} \\ \sigma_{RR} \\ \sigma_{Fly} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ -1 & 0.5 & 0.5 \end{pmatrix} \begin{pmatrix} \sigma_{ATM} \\ \sigma_{25\,call} \\ \sigma_{25\,put} \end{pmatrix}$$

$$\begin{pmatrix} \sigma_{ATM} \\ \sigma_{25\,call} \\ \sigma_{25\,put} \end{pmatrix} = \begin{pmatrix} \sigma_{ATM} \\ \sigma_{RR} \\ \sigma_{Fly} \end{pmatrix} A^{-1}$$

$$f'(x) \approx \frac{f(x+h) - f(x)}{h}; \quad h \text{ pequeña}$$

$$Vega = \frac{\partial C}{\partial \sigma} \approx \frac{C(\sigma + \Delta\sigma) - C}{\Delta\sigma}$$

$$Vanna = \frac{\partial^2 C}{\partial \sigma \partial S} \approx \frac{Vega(F + \Delta F) - Vega}{\Delta F}$$

$$Volga = \frac{\partial^2 C}{\partial \sigma \partial \sigma} \approx \frac{Vega(\sigma + \Delta\sigma) - vega}{\Delta\sigma}$$

constante

→ Black-Scholes: $dS_t = rS_t dt + \sigma S_t dW_t$

→ Local Vol: $dS_t = rS_t dt + \sigma(S_t, t) S_t dt$

Función determinista

Dependiente del nivel de Subyacente y del tiempo

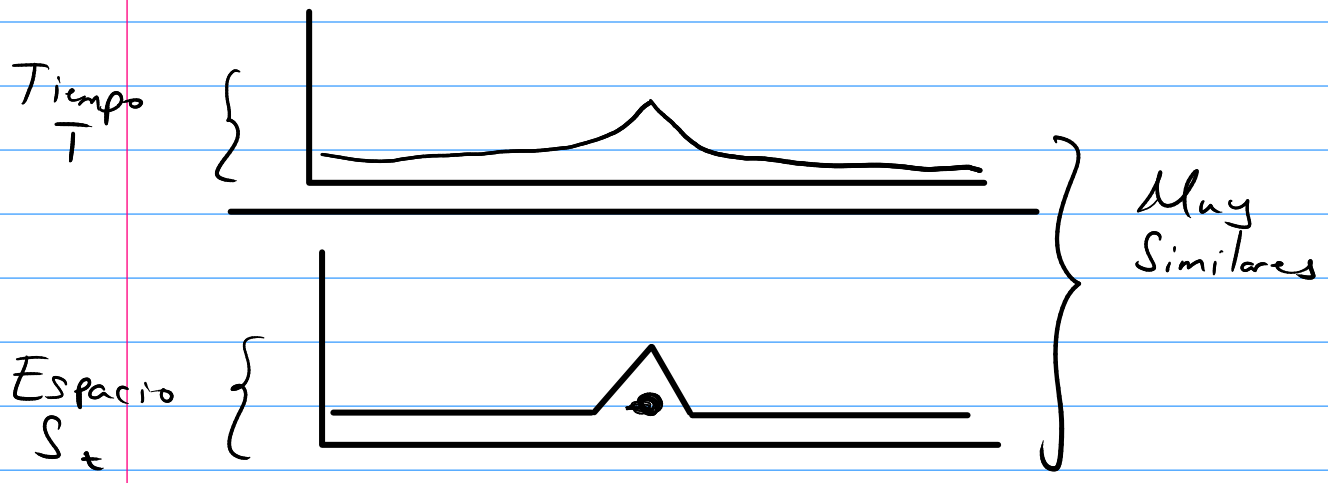
$$\frac{\frac{\partial C}{\partial T}}{\frac{\partial^2 C}{\partial K^2}} \rightarrow 0$$

($\frac{\partial C}{\partial T}$ → muy bajo; $\frac{\partial^2 C}{\partial K^2}$ → muy alto)

$$\frac{\partial C}{\partial T} = +1 \qquad \frac{\partial^2 C}{\partial K^2} = \Gamma$$

$$\frac{C(K+\Delta K) - 2C(K) + C(K-\Delta K)}{(\Delta K)^2}$$

$$\rightarrow \frac{C(T+\Delta T) - C(T)}{\Delta T}$$



Pour $\forall$ strike

$$dS_t = rS_t dt + \sigma S_t dW_t$$

$$C = C(S_t, T, \sigma(K,T))$$

$$\frac{\partial C}{\partial K} = \frac{\partial C^{BS}}{\partial \sigma} \cdot \frac{\partial \sigma(K,T)}{\partial K}$$ } Regle de la cadena

$$\frac{\partial^2 C}{\partial K^2} \quad ; \quad \frac{\partial C}{\partial T}$$

→ Socth - local Vol

local stoch.

$$\begin{cases} dS_t = rS_t dt + \sigma^*(S_t, t)\gamma(t) dW_t^{(1)} \\ dV_t = \alpha_t dt + \beta_t dW_t \end{cases}$$

- Jim Gatheral → Volatility Surfaces, a practitioner's guide.
- Emmanuel-Derman → Volatility Surfaces
- {Uwe Wysup ~ Volat. FX